

Brazilian E-Commerce Analytics

A Relational Database & Business Intelligence Pipeline

Mohammed S. Rahman

June 2026

Abstract

This report documents the end-to-end design, implementation, and visualisation of a relational analytics pipeline built on the Olist Brazilian E-Commerce public dataset. The pipeline spans database schema creation, data quality validation, the construction of reusable analytical views, and a suite of five Power BI dashboard pages covering executive KPIs, revenue trends, product performance, customer retention, and delivery & review quality. Across the dataset's approximately 100,000 orders placed between 2016 and 2018, total platform revenue stands at \$15.42 million from 96,000 delivered orders, with an average order value of \$159.83. Health & beauty and watches & gifts are the highest-revenue product categories, São Paulo sellers generate the dominant share of platform revenue, and the overall late-delivery rate of 8.11% is associated with a measurable decline in customer review scores. The repeat-customer rate of just 3% highlights a structural retention challenge that represents the platform's most significant operational opportunity for improvement.

Contents

1	Introduction	3
2	The Data	3
2.1	Dataset Overview	3
2.2	Schema Design	4
2.3	Data Quality Findings	5
2.3.1	Order Status Distribution	5
2.3.2	Missing Delivery Dates	5
2.3.3	Review Score Distribution	5
2.3.4	Payment Method Distribution	5
3	The Analytics Pipeline	5
3.1	Core Analytical Views (Script 03)	6
3.1.1	<code>vw_order_metrics</code>	6
3.1.2	<code>vw_category_revenue</code>	7
3.1.3	<code>vw_customer_retention</code>	7
3.2	Cohort Retention Analysis (Script 06)	7
3.3	Delivery–Review Relationship (Script 07)	8
3.4	Seller Revenue Concentration (Script 08)	8
3.5	Dashboard Views (Script 09)	8
3.6	Data Loading Utility	8
4	Power BI Dashboard	9
4.1	Executive Overview	9
4.1.1	Revenue and Order Volume Trends	10
4.1.2	Revenue by State	10
4.2	Product Analysis	10
4.2.1	Top Categories by Revenue	11
4.2.2	Top Categories by Items Sold	11
4.2.3	Revenue Share	11
4.2.4	Freight Cost Burden by Category	11
4.3	Customer Retention	12
4.3.1	One-Time vs Repeat Customers	12
4.3.2	New Customer Acquisition Trend	12
4.3.3	Orders per Customer Distribution	13
4.3.4	Cohort Retention Table	13
4.4	Delivery & Reviews	13
4.4.1	Review Score Distribution	13
4.4.2	Delivery Speed and Review Score	14
4.4.3	Late Delivery vs On-Time Review Scores	14
4.4.4	Late Delivery Rate by State	14
4.5	Seller Analysis	15
4.5.1	Top Sellers by Revenue	15
4.5.2	Seller Revenue by State	15
4.5.3	Revenue Concentration	15
4.5.4	Review Score vs Revenue Scatter	16

5	Discussion	16
5.1	Summary of Findings	16
5.2	Limitations and Future Work	17
6	Conclusion	17
A	SQL Script: Table Creation (01_create_tables.sql)	17
B	SQL Script: Customer Retention View (03_analytics_views.sql excerpt)	19
C	Python Data Loading Script (load_reviews.py)	20

1 Introduction

E-commerce platforms generate rich transactional records that, when organised and queried systematically, yield actionable insights across revenue, operations, product assortment, and customer experience. This project applies a structured analytics workflow to the Olist Brazilian E-Commerce dataset — a publicly available collection of anonymised marketplace orders — with the goal of building a reproducible, query-driven intelligence layer and a multi-page Power BI dashboard accessible to non-technical stakeholders.

The project addresses four core business questions:

1. **Revenue & Growth:** How has platform revenue evolved over time, which states drive the most orders, and what is the trajectory of average order value?
2. **Product Performance:** Which product categories generate the most revenue and volume, and where does freight cost create margin pressure?
3. **Customer Retention:** What share of customers make repeat purchases, and how do new-customer acquisition rates evolve across cohorts?
4. **Operational Quality:** How does delivery speed correlate with customer review scores, and which states and sellers exhibit the worst late-delivery rates?

The remainder of this report is structured as follows. Section 2 describes the dataset and the relational schema. Section 3 documents the SQL pipeline in detail, from schema creation through analytical views and dashboard views. Section 4 presents and interprets each of the five Power BI dashboard pages. Section 5 synthesises findings and identifies limitations. Section 6 concludes.

2 The Data

2.1 Dataset Overview

The Olist dataset is a Brazilian e-commerce public dataset made available on Kaggle, containing real commercial data from the Olist marketplace spanning September 2016 to August 2018. The dataset captures the full order lifecycle — from customer purchase through payment, dispatch, delivery, and post-delivery review — across multiple interlinked tables.

Table 1 summarises the approximate row counts for each table after loading and quality-checking the data.

Table 1: Approximate row counts per table after data loading.

Table	Row Count	Description
customers	99,441	One record per order-customer link
orders	99,441	Order header with status and timestamps
order_items	112,650	Line items with price and freight
order_payments	103,886	Payment records (multiple per order)
order_reviews	99,224	Post-delivery review scores and comments
products	32,951	Product attributes and dimensions
sellers	3,095	Seller location data
product_category_translation	71	Portuguese to English category names

2.2 Schema Design

The database uses a star-schema-style relational layout with `orders` as the central fact table. Figure 1 illustrates the entity relationships.

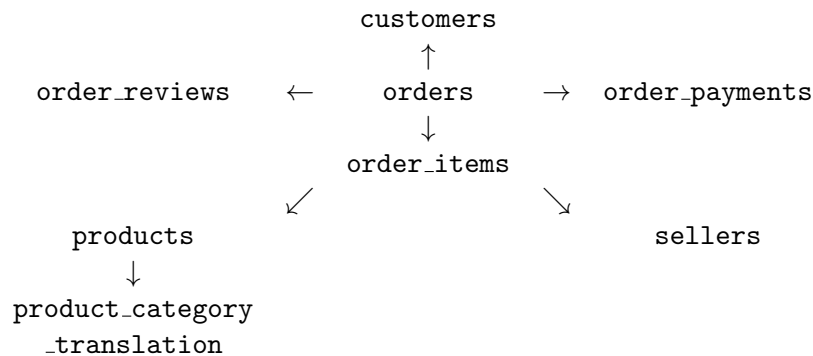


Figure 1: Conceptual entity-relationship diagram. `orders` sits at the centre, joined to `customers`, `order_items`, `order_payments`, and `order_reviews`. `order_items` links to both `products` and `sellers`.

The eight tables and their relationships are defined as follows:

- `customers` — stores one row per `customer_id` (which is order-scoped in this dataset), with a separate `customer_unique_id` used to identify the same buyer across multiple orders.
- `orders` — the primary fact table; each row is a single order with status and four key timestamps (purchase, approval, carrier handover, and customer delivery).
- `order_items` — a composite-keyed table (`order_id`, `order_item_id`) containing the price and freight value for every line item.
- `order_payments` — records one or more payment records per order, supporting split and instalment payments.
- `order_reviews` — one review per order, containing a 1–5 star score and optional comment fields.

- **products** — physical attributes including weight, dimensions, and Portuguese-language category names.
- **sellers** — geographic data (city, state, zip code prefix) for each seller.
- **product_category_translation** — a lookup table mapping Portuguese category names to English equivalents.

2.3 Data Quality Findings

Before analysis, a structured set of 15 data-quality checks was executed (SQL file `02_data_quality_checks.sql`). Key findings are summarised below.

2.3.1 Order Status Distribution

The vast majority of orders (approximately 96.5%) have a status of **delivered**, making the delivered-order subset the natural population for revenue and performance analysis. The remaining statuses include **shipped**, **canceled**, **unavailable**, **invoiced**, **processing**, and **created**.

2.3.2 Missing Delivery Dates

A meaningful proportion of orders lack a recorded `order_delivered_customer_date`, consistent with orders that did not reach the **delivered** status. All revenue and delivery performance metrics are therefore computed on the delivered subset only, using the `vw_order_metrics` view which filters explicitly to `order_status = 'delivered'`.

2.3.3 Review Score Distribution

The platform's review distribution is heavily right-skewed: 5-star reviews are the single largest group (approximately 57% of all reviews), while 1-star reviews constitute roughly 11%. The overall mean review score is 4.09 out of 5.

2.3.4 Payment Method Distribution

Credit card is the dominant payment method, accounting for approximately 74% of all payment records, followed by *boleto* (a Brazilian bank slip) at roughly 19%. Vouchers and debit cards make up the remainder.

3 The Analytics Pipeline

The analytics pipeline consists of nine sequential SQL scripts. Each script builds on the outputs of its predecessors, progressing from raw table creation through quality checks, analytical views, domain-specific analyses, and finally dashboard-ready views consumed directly by Power BI. Table 2 summarises each script's role.

Table 2: SQL pipeline script inventory.

Script	Purpose
01_create_tables.sql	Drops and recreates all eight base tables with foreign key constraints.
02_data_quality_checks.sql	15 diagnostic queries: row counts, status distributions, date coverage, review distributions, and join integrity checks.
03_analytics_views.sql	Creates three core analytical views reused throughout the pipeline: <code>vw_order_metrics</code> , <code>vw_category_revenue</code> , and <code>vw_customer_retention</code> .
04_revenue_analysis.sql	12 queries covering total revenue, monthly trends, state-level breakdowns, month-over-month growth, and AOV.
05_product_analysis.sql	12 queries on category revenue, volume, freight burden, revenue concentration, and review-weighted performance.
06_customer_retention_analysis.sql	12 queries on repeat-customer rates, cohort retention, first-purchase trends, and customer segmentation.
07_delivery_review_analysis.sql	16 queries on delivery speed, late rates, review score distributions, and the delivery-to-review relationship.
08_seller_analysis.sql	14 queries on seller revenue, concentration, review performance, late-delivery rates, and state-level seller metrics.
09_dashboard_views.sql	15 <code>CREATE OR REPLACE VIEW</code> statements producing the final aggregated views consumed by Power BI.

3.1 Core Analytical Views (Script 03)

Three foundational views are created in `03_analytics_views.sql` and referenced extensively by all downstream scripts.

3.1.1 *vw_order_metrics*

This is the primary analytical view, joining `orders`, `customers`, and `order_items` for all delivered orders. It computes, per order:

- **Revenue components:** `product_revenue`, `freight_revenue`, and `total_order_value`.
- **Delivery performance:** `delivery_days` (days from purchase to customer delivery) and `is_late_delivery` (a binary flag set to 1 when actual delivery exceeds the estimated delivery date).
- **Time dimensions:** `order_month` via `DATE_TRUNC('month', ...)` for monthly aggregation.

A representative excerpt of the view definition is shown below:

```
1 CREATE OR REPLACE VIEW vw_order_metrics AS
2 SELECT
```

```

3      o.order_id,
4      o.customer_id,
5      c.customer_unique_id,
6      c.customer_state,
7      DATE_TRUNC('month', o.order_purchase_timestamp) AS
order_month,
8      SUM(oi.price) AS product_revenue,
9      SUM(oi.freight_value) AS freight_revenue,
10     SUM(oi.price + oi.freight_value) AS total_order_value,
11     CASE
12         WHEN o.order_delivered_customer_date
13             > o.order_estimated_delivery_date
14         THEN 1 ELSE 0
15     END AS is_late_delivery,
16     EXTRACT(DAY FROM
17         o.order_delivered_customer_date
18         - o.order_purchase_timestamp) AS delivery_days
19 FROM orders o
20 JOIN customers c ON o.customer_id = c.customer_id
21 JOIN order_items oi ON o.order_id = oi.order_id
22 WHERE o.order_status = 'delivered'
23 GROUP BY o.order_id, o.customer_id, c.customer_unique_id,
24          c.customer_state, o.order_purchase_timestamp,
25          o.order_delivered_customer_date,
26          o.order_estimated_delivery_date;

```

Listing 1: Core delivered-order metrics view (excerpt).

3.1.2 *vw_category_revenue*

This view joins `order_items`, `products`, and `product_category_translation` to produce revenue, volume, and average price metrics per product category. The `COALESCE` pattern handles any categories that lack an English translation by falling back to the Portuguese name.

3.1.3 *vw_customer_retention*

Using a common table expression (CTE) with `ROW_NUMBER()` partitioned by `customer_unique_id`, this view assigns an order sequence number to each delivered order per unique customer. The outer query then aggregates to flag each customer as a repeat buyer (`is_repeat_customer = 1`) if they placed more than one delivered order.

3.2 Cohort Retention Analysis (Script 06)

The most technically involved query in the pipeline is the cohort retention analysis in `06_customer_retention_analysis.sql`. It uses a four-CTE chain to compute, for each monthly cohort, the share of customers still active at each subsequent month since first purchase. The structure is:

1. `customer_orders` — one row per (unique customer, order month) pair.

2. `cohorts` — the minimum order month for each unique customer, defining the cohort assignment.
3. `cohort_activity` — cross-joins customers to their cohort to compute `months_since_first_purchase` using `EXTRACT(YEAR)` and `EXTRACT(MONTH)` arithmetic.
4. `cohort_counts` and `cohort_sizes` — aggregate active customer counts, then divide by cohort size at month 0 to yield `retention_percent`.

3.3 Delivery–Review Relationship (Script 07)

Script 07 quantifies how delivery speed and on-time performance correlate with customer satisfaction. Three analytical approaches are used:

- **Late vs on-time review score:** joining `vw_order_metrics` with `order_reviews` and grouping by `is_late_delivery`.
- **Delivery speed buckets:** orders are bucketed into five groups (0–3 days, 4–7 days, 8–14 days, 15–30 days, 31+ days) and the average review score computed for each.
- **Review distribution by delivery status:** a window function computes the within-status percentage of each review score (1–5), enabling a side-by-side comparison of the full review distribution for late versus on-time orders.

3.4 Seller Revenue Concentration (Script 08)

The seller analysis identifies a concentration pattern in platform revenue. Using `RANK()` over seller-level revenue followed by a `CASE` bucketing step, the pipeline quantifies what fraction of total revenue is generated by the top 10, top 11–50, top 51–100, and all remaining sellers.

3.5 Dashboard Views (Script 09)

Fifteen `CREATE OR REPLACE VIEW` statements consolidate the analytical results into clean, aggregated views designed for direct Power BI connection via a PostgreSQL connector. Each view corresponds to one or more visualisations in the dashboard. Named consistently with a `dashboard_` prefix (e.g. `dashboard_executive_kpis`, `dashboard_cohort_retention`, `dashboard_seller_concentration`), these views pre-compute all aggregations so that Power BI can load data efficiently without performing complex joins at query time.

3.6 Data Loading Utility

The Python script `load_reviews.py` handles the loading of the order reviews CSV into PostgreSQL. The reviews file required special handling due to its `latin1` character encoding (common in Brazilian data exports). The script uses `pandas` with `sqlalchemy` and `psycopg2`, parsing the two timestamp columns (`review_creation_date` and `review_answer_timestamp`) with `pd.to_datetime(..., errors='coerce')` to gracefully handle any malformed date strings. Data is written to the `order_reviews` table in chunks of 1,000 rows using the `multi` insert method.

4 Power BI Dashboard

The Power BI dashboard comprises five pages, each connected to the corresponding `dashboard_*` views in PostgreSQL. The pages are: Executive Overview, Revenue (embedded in the Executive Overview), Product Analysis, Customer Retention, and Delivery & Reviews, and Seller Analysis.

4.1 Executive Overview



Figure 2: Executive Overview dashboard page showing top-line KPIs, monthly revenue and order trends (2016–2018), and revenue by customer state.

The Executive Overview page (Figure 2) presents six headline KPIs alongside monthly trend charts and a revenue-by-state breakdown. Key figures are summarised in Table 3.

Table 3: Executive KPI summary from `dashboard_executive_kpis`.

Metric	Value
Total Revenue	\$15.42 million
Total Delivered Orders	96,000
Average Order Value (AOV)	\$159.83
Average Review Score	4.09 / 5
Late Delivery Rate	8.11%
Repeat Customer Rate	3.00%

4.1.1 Revenue and Order Volume Trends

Monthly revenue grew substantially from near-zero in late 2016 to approximately \$8–9 million accumulated by mid-2018. The monthly orders chart exhibits a very similar shape, with a marked acceleration through 2017 that plateaus into 2018. This parallel growth in both order count and revenue indicates that the average order value remained relatively stable over the period, a finding confirmed by the AOV metric of \$159.83.

A notable feature of the revenue trend is the apparent levelling off in early-to-mid 2018. This could reflect seasonal effects, market saturation in the platform's existing customer base, or data truncation at the dataset's August 2018 endpoint.

4.1.2 Revenue by State

The state-level bar chart reveals high geographic concentration. São Paulo (SP) dominates by a considerable margin, consistent with its status as Brazil's most populous and economically active state. Rio de Janeiro, Minas Gerais, and other south/southeast states follow. Northern and northeastern states contribute minimally to platform revenue, which may reflect both lower internet penetration and longer delivery times to those regions — a theme reinforced by the Delivery & Reviews dashboard.

4.2 Product Analysis

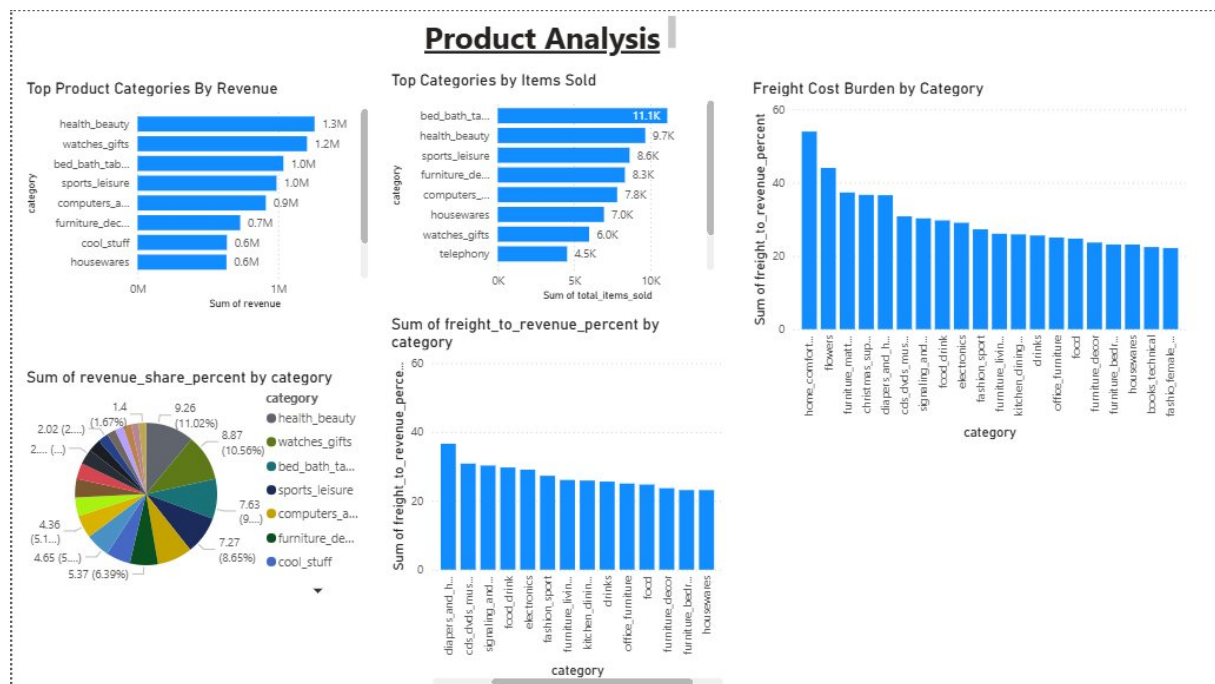


Figure 3: Product Analysis dashboard page showing top categories by revenue and items sold, revenue share, and freight cost burden.

The Product Analysis page (Figure 3) provides four views of category-level performance sourced from `vw_category_revenue` and `dashboard_product_categories`.

4.2.1 Top Categories by Revenue

Health & beauty leads all categories at approximately \$1.3 million in product revenue, followed closely by watches & gifts (\$1.2 million) and bed, bath & table (\$1.0 million). Sports & leisure and computers & accessories round out the top five, each contributing around \$0.9–1.0 million. The top five categories collectively account for roughly 45% of total platform revenue — a concentration level that is significant but not extreme, suggesting that the Olist marketplace has a moderately diversified category mix.

4.2.2 Top Categories by Items Sold

By volume (items sold), bed, bath & table takes the top position at approximately 11,100 items, followed by health & beauty (9,700 items) and sports & leisure (8,600 items). The divergence between the revenue and volume rankings is economically informative: watches & gifts ranks second in revenue but seventh in volume, implying a high average selling price, while bed, bath & table ranks first in volume but third in revenue, indicating lower-priced items sold in large quantities.

4.2.3 Revenue Share

The revenue share pie chart confirms category concentration: health & beauty holds approximately 11% of total revenue, watches & gifts around 10.6%, and the remaining categories each contribute less than 10%. The long tail of small categories is visually represented by the crowded outer portion of the chart, reinforcing that the revenue distribution is moderately concentrated rather than winner-takes-all.

4.2.4 Freight Cost Burden by Category

The freight-to-revenue percentage chart highlights a structurally important margin risk. The `home_comfort` and `flowers` categories exhibit freight costs exceeding 50% of product revenue — an economically unviable position for sellers or the platform. Bulky and fragile goods (furniture, large home items) unsurprisingly feature high freight burdens. Categories such as watches & gifts and computers show moderate freight ratios in the 20–35% range, reflecting their higher price points relative to weight and volume.

4.3 Customer Retention

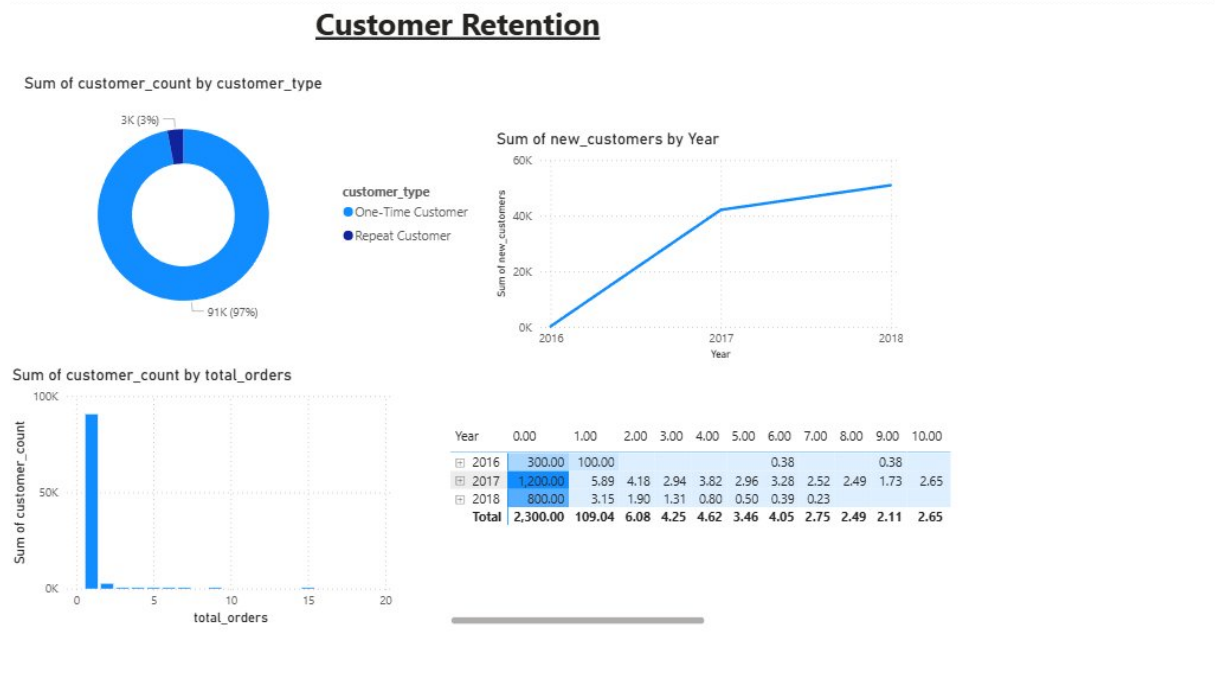


Figure 4: Customer Retention dashboard page showing one-time versus repeat customer breakdown, new customer acquisition over time, orders-per-customer distribution, and cohort retention table.

The Customer Retention page (Figure 4) is sourced from `vw_customer_retention`, `dashboard_customer_retention`, `dashboard_orders_per_customer`, `dashboard_new_customers_by_month`, and `dashboard_cohort_retention`.

4.3.1 One-Time vs Repeat Customers

The donut chart reveals a striking finding: 97% of customers (approximately 91,000) made only one delivered purchase, while just 3% (approximately 3,000) returned for a second or subsequent order. This near-complete absence of repeat purchasing behaviour is the most commercially significant finding in the dataset. It suggests that the platform is operating almost entirely as a new-customer acquisition engine rather than a retention platform — a fundamentally expensive model, given that customer acquisition typically costs significantly more than retention.

4.3.2 New Customer Acquisition Trend

The line chart of new customers by year shows rapid growth: near-zero in 2016, a steep rise through 2017 to approximately 40,000 new customers, and continued growth into 2018 approaching 50,000. This growth trajectory demonstrates strong top-of-funnel performance, but the retention data contextualises it: virtually none of those new customers returned.

4.3.3 Orders per Customer Distribution

The bar chart of customer count by total orders confirms the dominance of single-order customers, with the bar at `total_orders = 1` dwarfing all others by orders of magnitude. A small but visible tail extends to 5–10+ orders for a tiny segment of highly engaged customers, but these represent a negligible fraction of the base.

4.3.4 Cohort Retention Table

The cohort table (right side of Figure 4) displays the count of customers returning in each subsequent month after their first purchase, segmented by cohort year. The 2016 cohort (300 customers at month 0) shows marginal retention to months 1 and beyond. The 2017 cohort (1,200 at month 0) retains roughly 100 customers at month 1 (approximately 8%) and rapidly approaches zero by month 3. The 2018 cohort shows similar or weaker patterns. These figures are consistent with the headline 3% repeat-customer rate and confirm that retention is a persistent structural issue rather than a recent trend.

4.4 Delivery & Reviews

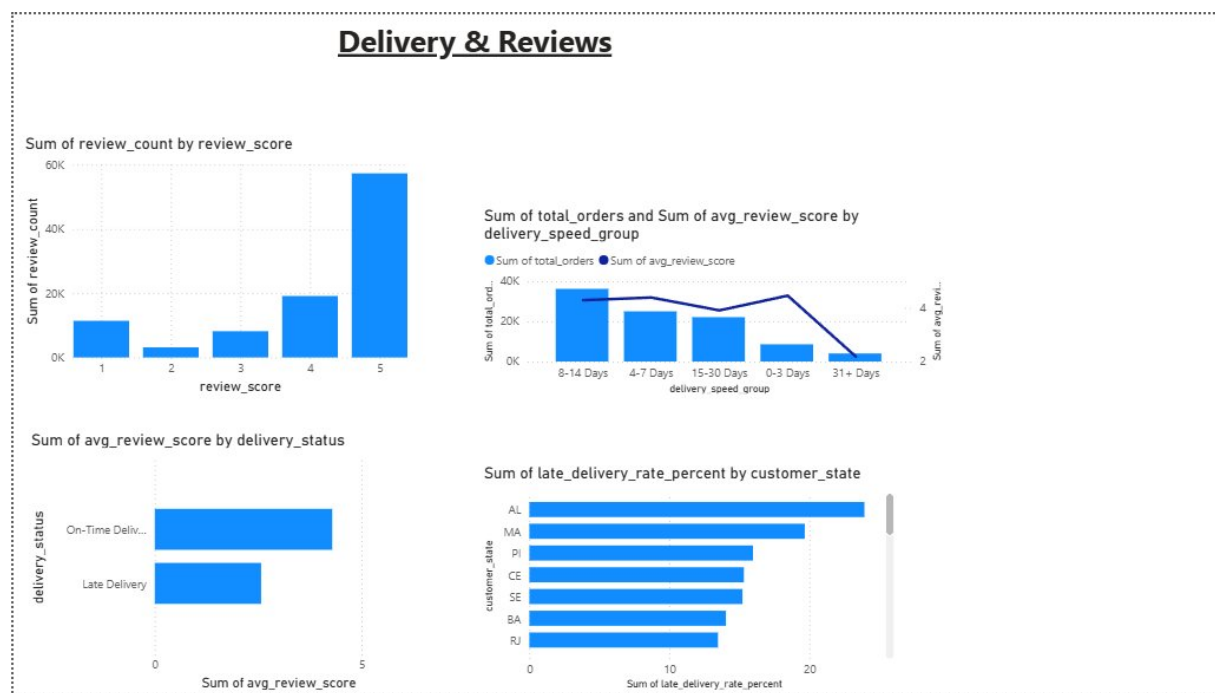


Figure 5: Delivery & Reviews dashboard page showing review score distribution, delivery speed vs. review score, delivery status impact on satisfaction, and late-delivery rate by customer state.

The Delivery & Reviews page (Figure 5) draws from `dashboard_delivery_reviews`, `dashboard_delivery_speed_groups`, `dashboard_delivery_by_state`, and `dashboard_review_distribution`.

4.4.1 Review Score Distribution

The histogram of review scores shows a strongly bimodal distribution: 5-star reviews dominate at approximately 57,000 reviews, with a secondary mode at 1-star (approximately

11,000 reviews). Scores 2, 3, and 4 are less frequent. This polarisation is a common pattern in marketplace review systems, where customers are most motivated to leave reviews at the extremes of experience.

4.4.2 *Delivery Speed and Review Score*

The combined bar-and-line chart (top-right of Figure 5) plots total orders by delivery speed group alongside the average review score for each group. Results confirm a clear positive relationship between faster delivery and higher satisfaction:

- Orders delivered in **0–3 days** attract the highest average review scores, approaching 4.4 out of 5.
- Orders in the **4–7 day** and **8–14 day** buckets show moderate scores around 4.1–4.2.
- Orders taking **15–30 days** see scores decline to approximately 3.8.
- Orders taking **31+ days** show the lowest satisfaction, with average scores around 2.5–2.8.

The volume distribution peaks in the 8–14 day bucket, which is the modal delivery experience on the platform. Reducing this to 4–7 days for a meaningful fraction of orders would likely produce a measurable improvement in the platform’s average review score.

4.4.3 *Late Delivery vs On-Time Review Scores*

The horizontal bar chart confirms that on-time deliveries receive a substantially higher average review score than late deliveries. On-time orders average approximately 4.3–4.4 stars, while late orders average roughly 2.7–2.9 stars — a difference of approximately 1.5 rating points. This gap has direct implications for seller reputation and platform trust.

4.4.4 *Late Delivery Rate by State*

The horizontal bar chart of late delivery rates by customer state (bottom-right of Figure 5) identifies Alagoas (AL), Maranhão (MA), and Piauí (PI) as the states with the highest proportions of late deliveries, with rates exceeding 20% for the worst performers. These states are predominantly in Brazil’s northeast region, consistent with longer last-mile distances from the São Paulo-concentrated seller base and less developed logistics infrastructure.

4.5 Seller Analysis



Figure 6: Seller Analysis dashboard page showing top sellers by revenue, seller revenue by state, revenue concentration, and seller review performance scatter plot.

The Seller Analysis page (Figure 6) draws from `dashboard_seller_performance`, `dashboard_seller_state` and `dashboard_seller_concentration`.

4.5.1 Top Sellers by Revenue

The top-ten sellers by revenue chart shows the leading seller generating approximately \$230,000 in product revenue, with a relatively steep drop-off to sellers 2–10, each contributing in the \$130,000–\$200,000 range. These are large individual operators by marketplace standards, likely representing either multi-category merchants or high-volume niche specialists.

4.5.2 Seller Revenue by State

São Paulo (SP) is the dominant seller state by a wide margin, with total seller revenue approaching \$10 million — more than double the combined contribution of all other states. Paraná (PR) and Minas Gerais (MG) are the next largest seller states. This geographic concentration of sellers in the south/southeast mirrors the customer concentration in the same region, and has implications for delivery times to customers in more distant states.

4.5.3 Revenue Concentration

The donut chart of seller revenue concentration reveals that the top 10 sellers account for approximately 19.9% of total platform revenue (from the 3,000+ active sellers). The top 11–50 sellers contribute a further 13.3%, and the top 51–100 add 12.3%, leaving approximately 54.5% of revenue generated by the long tail of smaller sellers. This is a

healthier concentration profile than many marketplaces, where a handful of sellers can dominate 70–80% of revenue.

4.5.4 *Review Score vs Revenue Scatter*

The scatter plot of average review score against seller revenue (colour-coded by state) shows a broadly positive cloud without strong linear structure: most sellers cluster at review scores between 3.5 and 4.5, with revenue varying widely within that band. A small number of high-revenue sellers show review scores below 3.0, representing a risk to platform quality. The colour coding reveals that São Paulo sellers (in purple/blue tones) populate both the high-revenue and the full range of review score positions.

5 Discussion

5.1 Summary of Findings

This project surfaces five findings of primary commercial importance.

1. Revenue growth is strong but decelerating. Platform revenue grew steeply through 2017 and into early 2018 before showing signs of deceleration. The average order value of \$159.83 has remained stable, indicating that growth was volume-driven rather than ticket-size-driven. Sustaining revenue growth will require either continued new-customer acquisition or — more efficiently — improvements to the repeat-purchase rate.

2. Customer retention is critically low. A 3% repeat-customer rate is the most important single finding in the dataset. The economic implications are severe: every dollar of revenue growth requires almost exactly one dollar of customer acquisition spend, and the platform captures almost no benefit from the compounding loyalty that characterises healthy marketplaces. Initiatives to increase repeat purchasing — loyalty programmes, post-purchase re-engagement, subscription-style offers — represent the highest-leverage investment available.

3. Delivery speed is the dominant driver of customer satisfaction. The 1.5-star average review score gap between late and on-time deliveries, and the monotone decline in scores across speed buckets, establishes delivery performance as the most operationally actionable lever for improving customer experience. With 8.11% of orders arriving late, and northeastern states suffering rates above 20%, there is meaningful headroom for improvement.

4. Product concentration is moderate and healthy. The top five categories account for roughly 45% of revenue, and no single category dominates. This diversification mitigates category risk and suggests the platform has successfully recruited merchants across a broad assortment. The freight burden analysis, however, flags specific categories (home comfort, flowers, and oversized goods) where high shipping costs likely suppress seller viability or require subsidisation.

5. Seller geography creates a structural logistics challenge. The concentration of both sellers and customers in São Paulo and adjacent states, combined with the low seller presence in the north and northeast, creates systematically longer and less reliable

delivery routes to those regions. Recruiting sellers in underserved states or establishing regional fulfilment infrastructure would be the most direct remedies.

5.2 Limitations and Future Work

Causal claims are not warranted. The analyses in this project are descriptive and associative. The observed correlation between late delivery and low review scores does not rule out the possibility that both are caused by a third factor (e.g., seller quality). Causal identification would require experimental variation in delivery conditions.

The 3% repeat-customer rate may be partially an artefact of the dataset. The Olist dataset's `customer_id` is order-scoped, meaning that a returning customer may receive a new `customer_id` if they place an order under slightly different account details. The `customer_unique_id` field is designed to mitigate this, but its completeness and accuracy is not formally documented. The true repeat-customer rate may be modestly higher than 3%.

The dataset ends in August 2018. Post-2018 growth, competitive dynamics, and any operational changes Olist made in response to early data signals are outside the scope of this analysis.

Predictive modelling is a natural extension. The structured database and analytical views created here provide a clean foundation for supervised learning applications — for example, predicting which customers are likely to make a second purchase (a counterpart to the churn prediction pipeline documented separately), or predicting delivery delay at the order level to trigger proactive customer communications.

6 Conclusion

This project delivers a complete analytics pipeline from raw CSV data to an interactive multi-page Power BI dashboard, grounded in a well-structured PostgreSQL relational schema and a layered SQL architecture. The pipeline produces reproducible, reusable analytical views that abstract complexity away from the dashboard layer, making future analytical queries and dashboard extensions straightforward.

The most commercially significant findings are the near-absence of repeat purchasing behaviour (3% repeat-customer rate), the strong link between delivery timeliness and customer satisfaction, and the geographic concentration of sellers in São Paulo that creates structural delivery disadvantages for northeastern customers. Together, these findings point toward a clear strategic agenda: invest in customer retention mechanisms, reduce late-delivery rates in underserved regions, and monitor high-revenue sellers with sub-par review performance before reputational damage compounds.

The SQL scripts, Python data-loading utility, and Power BI dashboard represent a self-contained, extensible artefact that could be maintained and updated as new monthly order data becomes available. The dashboard views are designed for direct PostgreSQL connector refresh in Power BI, requiring no manual data transformation steps once the underlying tables are updated.

A SQL Script: Table Creation (01_create_tables.sql)

```
1 DROP TABLE IF EXISTS order_reviews;
2 DROP TABLE IF EXISTS order_payments;
3 DROP TABLE IF EXISTS order_items;
4 DROP TABLE IF EXISTS orders;
5 DROP TABLE IF EXISTS products;
6 DROP TABLE IF EXISTS sellers;
7 DROP TABLE IF EXISTS customers;
8 DROP TABLE IF EXISTS product_category_translation;
9
10 CREATE TABLE customers (
11     customer_id TEXT PRIMARY KEY,
12     customer_unique_id TEXT,
13     customer_zip_code_prefix INTEGER,
14     customer_city TEXT,
15     customer_state TEXT
16 );
17
18 CREATE TABLE sellers (
19     seller_id TEXT PRIMARY KEY,
20     seller_zip_code_prefix INTEGER,
21     seller_city TEXT,
22     seller_state TEXT
23 );
24
25 CREATE TABLE products (
26     product_id TEXT PRIMARY KEY,
27     product_category_name TEXT,
28     product_name_lenght FLOAT,
29     product_description_lenght FLOAT,
30     product_photos_qty FLOAT,
31     product_weight_g FLOAT,
32     product_length_cm FLOAT,
33     product_height_cm FLOAT,
34     product_width_cm FLOAT
35 );
36
37 CREATE TABLE orders (
38     order_id TEXT PRIMARY KEY,
39     customer_id TEXT,
40     order_status TEXT,
41     order_purchase_timestamp TIMESTAMP,
42     order_approved_at TIMESTAMP,
43     order_delivered_carrier_date TIMESTAMP,
44     order_delivered_customer_date TIMESTAMP,
45     order_estimated_delivery_date TIMESTAMP,
46     FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
47 );
48
49 CREATE TABLE order_items (
50     order_id TEXT,
51     order_item_id INTEGER,
```

```

52     product_id      TEXT,
53     seller_id       TEXT,
54     shipping_limit_date  TIMESTAMP,
55     price            NUMERIC(10, 2),
56     freight_value    NUMERIC(10, 2),
57     PRIMARY KEY (order_id, order_item_id),
58     FOREIGN KEY (order_id) REFERENCES orders(order_id),
59     FOREIGN KEY (product_id) REFERENCES products(product_id),
60     FOREIGN KEY (seller_id) REFERENCES sellers(seller_id)
61 );
62
63 CREATE TABLE order_payments (
64     order_id          TEXT,
65     payment_sequential INTEGER,
66     payment_type       TEXT,
67     payment_installments INTEGER,
68     payment_value      NUMERIC(10, 2),
69     FOREIGN KEY (order_id) REFERENCES orders(order_id)
70 );
71
72 CREATE TABLE order_reviews (
73     review_id          TEXT PRIMARY KEY,
74     order_id           TEXT,
75     review_score        INTEGER,
76     review_comment_title TEXT,
77     review_comment_message TEXT,
78     review_creation_date  TIMESTAMP,
79     review_answer_timestamp TIMESTAMP,
80     FOREIGN KEY (order_id) REFERENCES orders(order_id)
81 );
82
83 CREATE TABLE product_category_translation (
84     product_category_name      TEXT PRIMARY KEY,
85     product_category_name_english TEXT
86 );

```

Listing 2: Full table creation script.

B SQL Script: Customer Retention View (03_analytics_views.sql excerpt)

```

1 CREATE OR REPLACE VIEW vw_customer_retention AS
2 WITH customer_orders AS (
3     SELECT
4         c.customer_unique_id,
5         o.order_id,
6         o.order_purchase_timestamp,
7         ROW_NUMBER() OVER (
8             PARTITION BY c.customer_unique_id
9             ORDER BY o.order_purchase_timestamp
10        ) AS order_number

```

```

11     FROM orders o
12     JOIN customers c ON o.customer_id = c.customer_id
13     WHERE o.order_status = 'delivered'
14 )
15 SELECT
16     customer_unique_id,
17     COUNT(order_id)                AS total_orders,
18     MIN(order_purchase_timestamp) AS first_order_date,
19     MAX(order_purchase_timestamp) AS last_order_date,
20     CASE
21         WHEN COUNT(order_id) > 1 THEN 1
22         ELSE 0
23     END AS is_repeat_customer
24 FROM customer_orders
25 GROUP BY customer_unique_id;

```

Listing 3: Customer retention view using ROW_NUMBER() to identify repeat purchasers.

C Python Data Loading Script (load_reviews.py)

```

1 import pandas as pd
2 from sqlalchemy import create_engine
3 from urllib.parse import quote_plus
4
5 DB_USER      = "****"
6 DB_PASSWORD  = "****"
7 DB_HOST      = "localhost"
8 DB_PORT      = "****"
9 DB_NAME      = "ecommerce_analytics"
10
11 csv_path = r"C:\Users\..."
12 encoded_password = quote_plus(DB_PASSWORD)
13
14 engine = create_engine(
15     f"postgresql+psycopg2://{DB_USER}:{encoded_password}@{DB_HOST}:{DB_PORT}/{DB_NAME}"
16 )
17
18 df = pd.read_csv(csv_path, encoding="latin1", dtype=str)
19
20 date_cols = ["review_creation_date", "review_answer_timestamp"]
21 for col in date_cols:
22     df[col] = pd.to_datetime(df[col], errors="coerce")
23
24 df.to_sql(
25     "order_reviews", engine,
26     if_exists="append", index=False,
27     chunksize=1000, method="multi"
28 )
29
30 print("Loaded order_reviews successfully.")

```

Listing 4: Python script for loading the order reviews CSV into PostgreSQL, handling Latin-1 encoding and timestamp parsing.